Exam 1 Review (up to 5 points extra credit)

Due to Gradescope by 1:00 PM on Wednesday, October 12th

This extra credit opportunity cannot be submitted late

You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade.

Homework Goal: Practice the material we've covered up until Exam 1 and bring questions and confusions to class on Wednesday, October 12th

NOTE: The exam will be on Chapters 1-4, part of Chapter 5 (zyBooks 5.1 and 5.6), and Chapter 6. In addition to problems like the ones below, the exam will have true/false, fill in the blank, and multiple choice. You would be wise to study all the terminology we have learned in the class thus far; a set of definitions are available on Knowt.io (link is posted on Piazza). Also, homework solutions are posted on Canvas under Files.

Chapter 1/5: Introduction and Algorithms

1. Write pseudocode that computes a running sum of numbers that the user enters. When they enter a negative number, stop adding the numbers and print out the total sum.

x: true
sum=0
While x
num = get user input
if num is negative
x = false
else
sum += num
Print(num)

2. Write the output from the following pseudocode:

```
set count = 0
while count is NOT 7
   count = count + 2
   print count
```

2, 4, 6

3. Suppose the following sorted numbers exist in some list and you are looking for 403 via the binary search algorithm. What numbers would 403 be compared with to determine whether it is in the list or not (look left if you have a tie)?
[101, 102, 200, 220, 250, 262, 274, 306, 341, 358, 400, 403, 406, 442, 470, 471, 475]

1, 2, 3, 4

4 - searches

It would be compared with 400

Chapter 2: Binary

4. Complete the following conversions table:

Decimal	Binary (unsigned)	Base 5	Hex
12	0000 1100	22	C
37	0010 0101	122	25
165	1010 0101	1130	A5
209	1101 0001	1314	D1
63	0011 1111	223	3F

5. You have the following 8-bit number: 1100 0110. What is its decimal representation in each of the following notations:
   (a) Unsigned binary notation?

$2^7+2^6+2^2+2^1$

$-2^7+2^6+2^2+2^1$

198

   (b) Two's complement notation?

−58

6. You know that your machine stores a number using 5 bits. What is the range of numbers you can represent using:
   (a) Unsigned binary notation?

0 - 31

   (b) Two's complement notation?

-16 - 15

7. Assume your computer uses 8-bit 2's complement signed numbers. Perform the following additions, giving your answer in both binary and decimal. Indicate if overflow occurs.
   (a)

```
  0 0 1 0 1 0 1 1   43
+ 0 1 0 0 1 0 0 1   73
```

0111 0100

116

No overflow

(b)

```
  1 0 1 0 0 1 1 0   -90
+ 1 1 1 1 0 0 1 0   -14
```

1 1001 1000

-104 No overflow

8. What color is this RGB pixel? 11111111 11111111 00000000

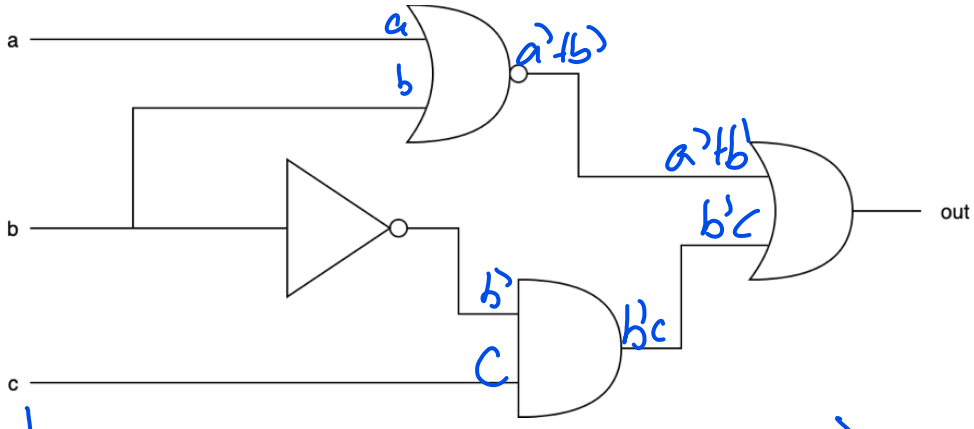
(255, 255, 0)

9. Here is an ASCII message. What does it say? (If we ask you to translate ASCII on the exam, you will be provided with a table.)
01000111 01101111 01101111 01100100 00100000 01101100 01110101 01100011 01101011 00100001

Good Luck!

Chapter 3: Boolean Expressions, Truth Tables, Circuits

10. Given the following circuit diagram, create a truth table and Boolean expression.



a	b	C	out
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

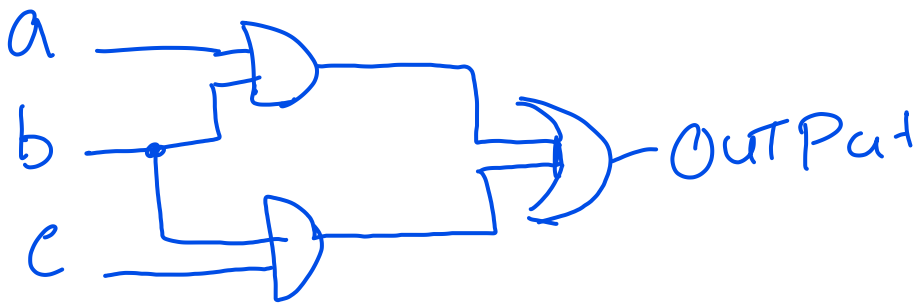
(a'+b') + (b'c)

11. Given the following truth table, create a Boolean expression and circuit.

a	b	c	output	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$a'bc$
1	0	0	0	
1	0	1	0	
1	1	0	1	abc'
1	1	1	1	abc

$a'bc + abc' + abc$

$(ab) + (bc)$



Chapter 3: Hardware

12. RAM addresses (Remember to use 2^n not 10^m):
    (a) If you have 11 bits available for RAM addresses, how many memory cells can theoretically exist?

2^{11}

$= 2048$

    (b) If you have 8 MiB of memory, then how many bits are needed to represent all memory addresses?

$8\ \text{MiB} \cdot \frac{2^{20}\text{B}}{1\ \text{MiB}} \cdot \frac{2^3\text{b}}{1\text{B}} = 67108864\text{bits}$

13. Suppose Alexi's digital camera has a storage capacity of 4GiB. What is the maximum number of photographs he can store in his camera if each picture is 1200 by 1800 pixels and each pixel requires 3 bytes of storage (RGB)?

$4\ \text{GiB} \cdot \frac{2^{30}}{1\ \text{GiB}} = 4294967296\text{B}$

$1\ \text{Photo} = 1200 \cdot 1800 \cdot 3$

$= 6480000\text{B}$

$\# = \frac{4294967296}{6480000}$

$= 662\ \text{Photos}$

14. Instruction Types
    (a) Suppose Abdul designed a machine language with an opcode field of 5 bits. How many different instruction types can Abdul's language contain?

2^5

$= 32$ opcodes

    (b) Lola designed a system with 7 bits for the opcode. How many instructions does her language support?

2^7

$= 128$ opcodes

Chapter 4: Software

15. List and describe the parts of an assembly language instruction.

opcode destination address address 1 address 2

Use the following table for questions (16) and (17).

Code	Assembly	Operation
00	sqrt	Square root
01	div	Division
10	add	Addition
11	sub	Subtraction

M[4] = a
M[5] = b
M[C] = C
M[7] = x

16. Write a program in assembly to compute $c = \sqrt{\frac{a-b}{a+b}}$. The addresses of a, b, c are M[4], M[5], and M[6], respectively. Also, assume M[7] is available for any intermediate answer.

11 M[7] M[4] M[5]
10 M[5] M[8] M[4]
01 M[4] M[7] M[5]
00 M[C] M[4] 000

17. Convert your assembly program from (16) into machine language. Use 2-bit opcodes and 3 bit addresses.

4 → 100 ; 5 → 101 ; 6 → 110 ; 7 → 111

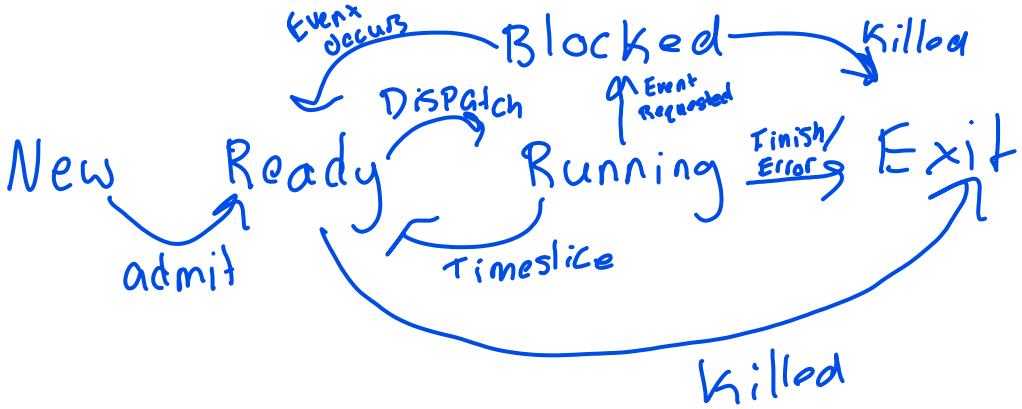
11 111 100 101
10 101 101 100
01 100 111 101
00 110 100 000

Chapter 6: Operating Systems

18. List the five process states discussed in class and define each with one sentence.

Ready - process that could run; in ready queue
Running - a process currently being executed
Blocked - process waiting for an event to occur
New - process being created by the OS
Exit - OS is cleaning up after terminated process

19. Draw the graph that shows all the labeled process transitions between the states.



20. Define the difference between an I/O intensive and a compute intensive process. What does the OS do to ensure a compute intensive process does not "hog" the system?

A CPU intensive process spends most time in running or ready state because it needs to use system resources but gets booted out from running because of Timeslice.
An I/O process spends time in blocked because it is waiting for an event or input to occur.

21. Suppose the overhead to swap a RUNNING process with a READY process is 1 ms. Is a time slice of 2 ms a good time slice amount? Justify your response.

No because then the process only gets about 1ms of running time before it has to switch and that is not a lot of time

22. Each process in a system spends 75% of its time in the BLOCKED state.

(a) If we are running 5 processes at once, what is the processor utilization in this system?

$1-(0.75)^5$

$= 76.3\%$

(b) How many processes should be run to guarantee 90% processor utilization?

$1-(0.75)^x = 0.9$

$x = 8$ processes

23. What is the difference between a physical address and a virtual address? Can they ever be the same number?

a physical address represents the physical location of the data in memory

a virtual address is more of a refrence or name for the physical address location and does not exist physically in memory

Yes, physical address and virtual address can be the same number but can also not be.

Physical Address	Virtual Address	Allocated:
0-15	0-15	Program A ← same
16-31	0-15	Program B ← not same